

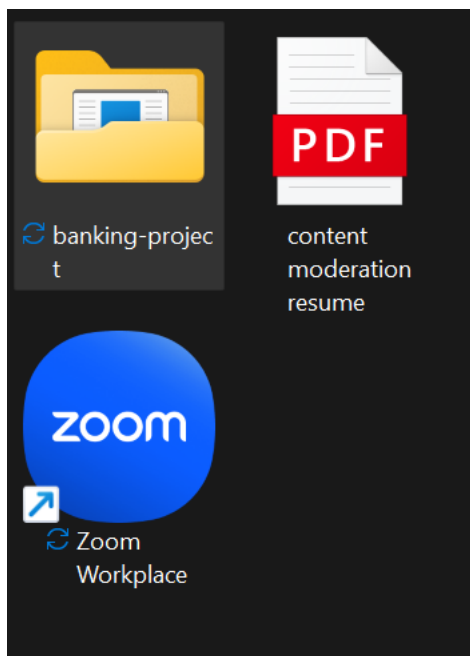
Finance Me Banking and Finance Domain

Certification Project – Finance Me Banking and Finance Domain

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Submitted to-Vikul Date of submission-18/1/2025

Step-1 Make Folder in my laptop Name “banking-project” and paste downloaded zip file in this folder



Step-2 Then open VS code and write terraform script . Created two file “authentication.tf” and “ec2.tf” for creating two instance

A screenshot of the Visual Studio Code editor. The left sidebar shows the Explorer view with a folder named 'BANKING-PROJECT'. Inside the folder, there are files: '.terraform', '.terraform.lock.hcl', 'ec2.tf' (selected), 'terraform.exe', and 'terraform.tfstate'. The main editor area shows the content of 'ec2.tf'. The script defines the AWS provider and creates two EC2 instances. The code is as follows:

```
1 # Define the provider with AWS credentials
2 provider "aws" {
3   region = "ap-south-1"
4   access_key = "AKIA2UC3BD7OTFD3VIMP"
5   secret_key = "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"
6 }
7
8 # Create two EC2 instances
9 resource "aws_instance" "ec2_instances" {
10   count = "2"
11   ami = "ami-00bb6a80f01f03502"
12   instance_type = "t3.medium"
13   key_name = "health-project"
14 }
15
```

Step-3 Run Terraform "init , plan or apply" command

```
8 # Create two EC2 instances
9 resource "aws_instance" "ec2_instances" {
10     count          = "2"

```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

Microsoft Windows [Version 10.0.22631.4751]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dell\OneDrive\Desktop\banking-project>terraform init

Initializing the backend...

Initializing provider plugins...

- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v5.84.0...
- Installed hashicorp/aws v5.84.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider selections it made above. Include this file in your version control repository so that Terraform can guarantee to make the same selections by default when you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

```
9 resource "aws_instance" "ec2_instances" {
10     count          = "2"

```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

on ec2.tf line 11, in resource "aws_instance" "ec2_instances":
11: ami = data.aws_ami.ubuntu.id

A data resource "aws_ami" "ubuntu" has not been declared in the root module.

C:\Users\dell\OneDrive\Desktop\banking-project>terraform plan

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

- + create

Terraform will perform the following actions:

```
# aws_instance.ec2_instances[0] will be created
+ resource "aws_instance" "ec2_instances" {
+   ami           = "ami-00bb6a80f01f03502"
+   arn           = (known after apply)
```

Note: You didn't use the `-out` option to save this plan, so Terraform can't guarantee to take exactly these actions if you run `"terraform apply"` now.

```
C:\Users\dell\OneDrive\Desktop\banking-project>terraform apply
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# aws_instance.ec2_instances[0] will be created
+ resource "aws_instance" "ec2_instances" {
  + ami           = "ami-00bb6a80f01f03502"
  + arn           = (known after apply)
```

Successfully two instances launch

Instances (1/2)

Info

Last updated less than a minute ago

Connect

Instance state ▾

Actions ▾

Launch instances ▾

Find Instance by attribute or tag (case-sensitive)

All states ▾

<

1

>

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
	banking-project	i-03087f167fd4a667b	Running	t3.medium	3/3 checks passed	View alarms +
	banking-test	i-037373e41b9114a5b	Running	t3.medium	3/3 checks passed	View alarms +

Step-4 Connect first machine with SSH connection and run command “`sudo su`” or “`apt update -y`” install git ,maven and Jenkins in the machine

```
root@ip-172-31-15-19:/home/ubuntu# history
1  apt update -y
2  sudo apt install -y git
3  git --version
4  sudo apt install -y maven
5  mvn -version
6  wget https://raw.githubusercontent.com/amanpandey8120/Deployment-script/refs/heads/main/jenkins.sh
7  ls
8  chmod 700 jenkins.sh
9  ./jenkins.sh
10 history
root@ip-172-31-15-19:/home/ubuntu#
```

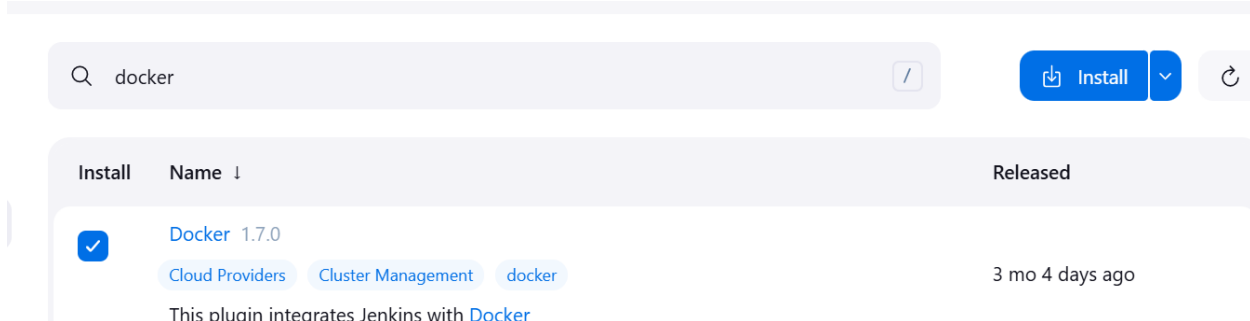
i-03087f167fd4a667b (banking-project)

PublicIPs: 13.232.205.176 PrivateIPs: 172.31.15.19

Step-5 Access my Jenkins with public ip



Step-6 In my Jenkins install docker plugin



Step-7 Install docker.io and run command “sudo usermod -aG docker Jenkins” and restarted Jenkins

```
11 cat /var/lib/jenkins/secrets/initialAdminPassword
12 apt install -y docker.io
13 sudo usermod -a -G docker Jenkins
14 sudo usermod -aG docker jenkins
15 service jenkins restart
16 history
root@ip-172-31-15-19:/home/ubuntu#
```

i-03087f167fd4a667b (banking-project)

PublicIPs: 13.232.205.176 PrivateIPs: 172.31.15.19

Step-8 After restarting the Jenkins again login and go to new item and start the banking-project

New Item

Enter an item name

banking-project

Select an item type



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



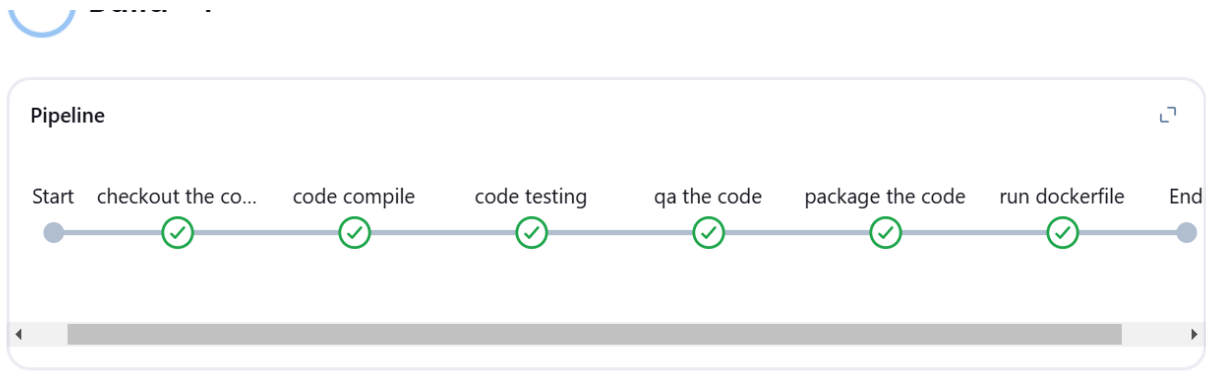
Pipeline

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.

```
1 pipeline {
2   agent any
3   stages {
4     stage('checkout the code from github') {
5       steps {
6         git url: 'https://github.com/amanpandey8120/star-agile-banking-finance.git'
7         echo 'github url checkout'
8       }
9     }
10    stage('code compile ') {
11      steps {
12        echo 'starting compiling'
13        sh 'mvn compile'
14      }
15    }
16    stage('code testing ') {
17      steps {
18        sh 'mvn test'
19      }
20    }
21    stage('qa the code') {
22      steps {
23        sh 'mvn checkstyle:checkstyle'
24      }
25    }
26    stage('package the code') {
27      steps {
28        sh 'mvn package'
29      }
30    }
31    stage('run dockerfile') {
32      steps {
33        sh 'docker build -t finance-me1 .'
34      }
35    }
36  }
37 }
```

try sample Pipeline... ▾

Write the jenkins file apply , save and build it



Step-9 Push the image to docker-hub with using CLI

```
17 docker images
18 docker login
19 docker tag finance-mel amanpandey123/finance-mel:latest
20 docker push amanpandey123/finance-mel:latest
21 history
root@ip-172-31-15-19:/home/ubuntu#
```

i-03087f167fd4a667b (banking-project)

PublicIPs: 13.232.205.176 PrivateIPs: 172.31.15.19

Step-10 Now we will execute the ansible-playbook

Go to manage Jenkins/plugins/available plugins install ansible plugins

Install

Name ↓

Released

☒	Ansible 403.v8d0ca_dcb_b_502 pipeline External Site/Tool Integrations DevOps Build Tools Deployment Invoke Ansible Ad-Hoc commands and playbooks. This plugin is up for adoption! We are looking for new maintainers. Visit our Adopt a Plugin initiative for more information.	7 mo 3 days ago
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Step-11 Install ansible in machine and add ansible-group in hosts file and restart the jenkins

```
21 history
22 sudo apt upgrade -y
23 apt-get install -y software-properties-common
24 apt-add-repository ppa:ansible/ansible
25 apt-get update
26 apt-get install -y ansible
27 ansible --version
28 vi /etc/ansible/hosts
29 service jenkins restart
30 history
root@ip-172-31-15-19:/home/ubuntu#
```

i-03087f167fd4a667b (banking-project)

PublicIPs: 13.232.205.176 PrivateIPs: 172.31.15.19

Step-12 Fill all the details and generate pipeline syntax and add line in the same Jenkins file for execution of the ansible playbook

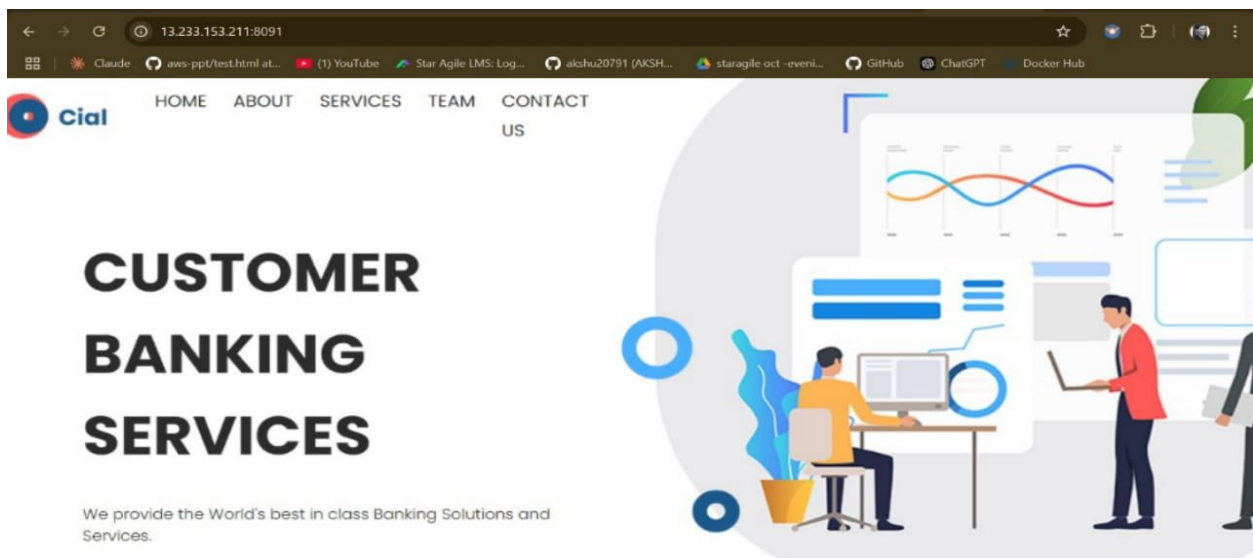
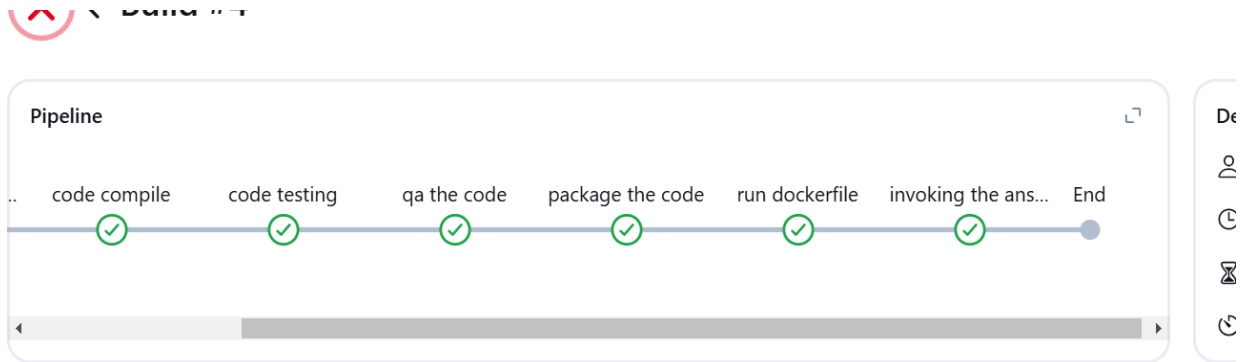
Generate Pipeline Script

```
ansiblePlaybook become: true, credentialsId: 'anssible', installation: 'ansible', inventory: '/etc/ansible/hosts', playbook: 'ansible-playbook.yml', vaultTmpPath: "
```



```
31 stage( run docketfile ) {
32     steps {
33         sh 'docker build -t finance-me1 .'
34     }
35 }
36 stage('invoking the ansible-playbook') {
37     steps {
38         ansiblePlaybook become: true, credentialsId: 'anssible', installation: 'ansible', inventory: '/'
39     }
40 }
41 }
42
```

☒ Use Groovy Sandbox ?



Step-13 Now Launch one more Instance name “banking monitor”

Find Instance by attribute or tag (case-sensitive)							
Running							
	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	
☒	banking-test	i-037373e41b9114a5b	Running	t3.medium	3/3 checks passed	View alarms	+
☐	banking-monitor	i-013e3a2e8e4453e3a	Running	t2.micro	2/2 checks passed	View alarms	+
☐	banking -project	i-08c71592ac983d1da	Running	t3.medium	3/3 checks passed	View alarms	+

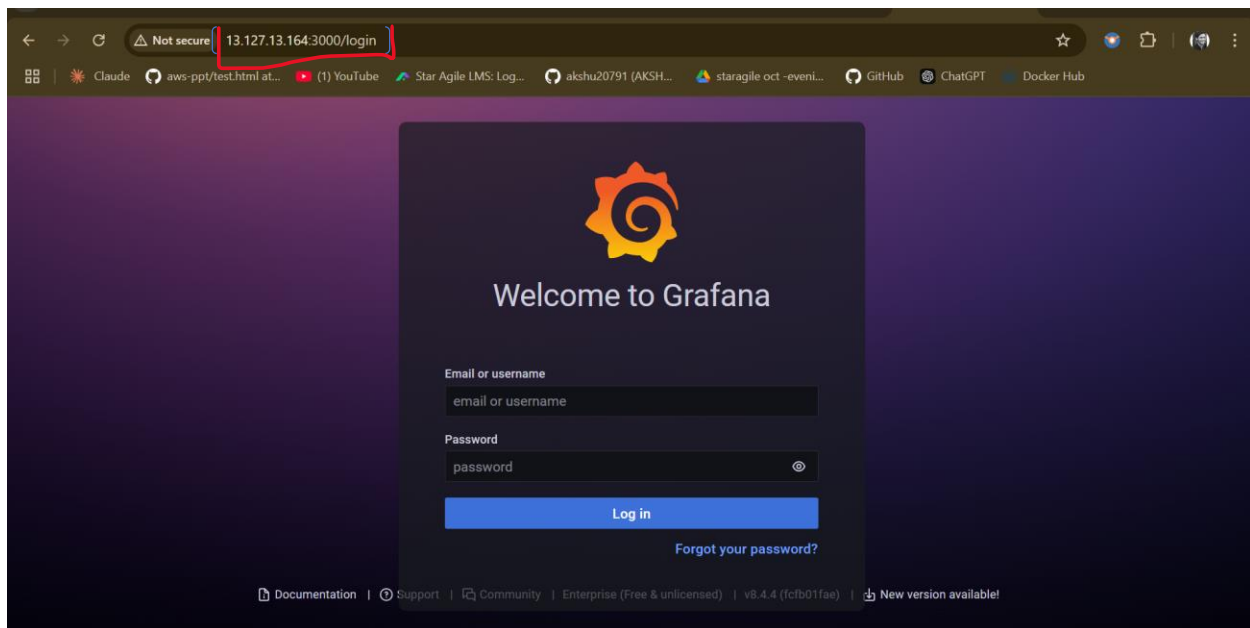
Step-14 Install Grafana in monitor machine


```
Last login: Fri Jan 17 14:22:43 2025 from 13.233.177.5
ubuntu@ip-172-31-1-128:~$ sudo su
root@ip-172-31-1-128:/home/ubuntu# history
 1  wget https://dl.grafana.com/enterprise/release/grafana-enterprise-8.4.4.linux-amd64.tar.gz
 2  history
root@ip-172-31-1-128:/home/ubuntu#
```

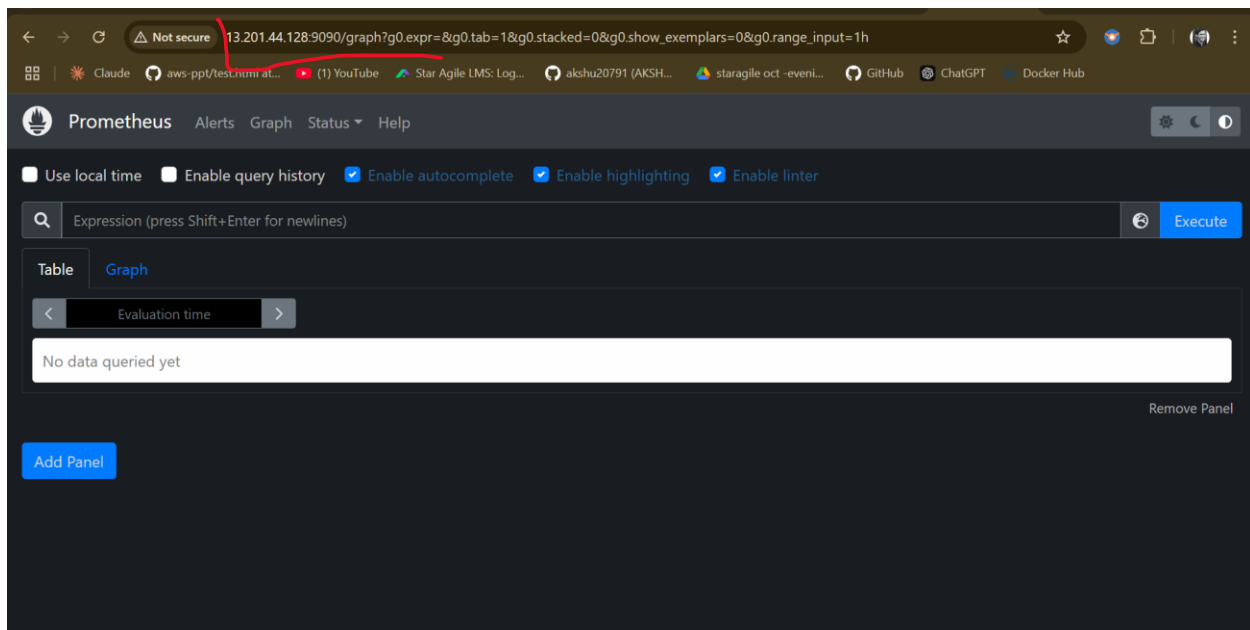
i-013e3a2e8e4453e3a (banking-monitor)

PublicIPs: 13.127.13.164 PrivateIPs: 172.31.1.128

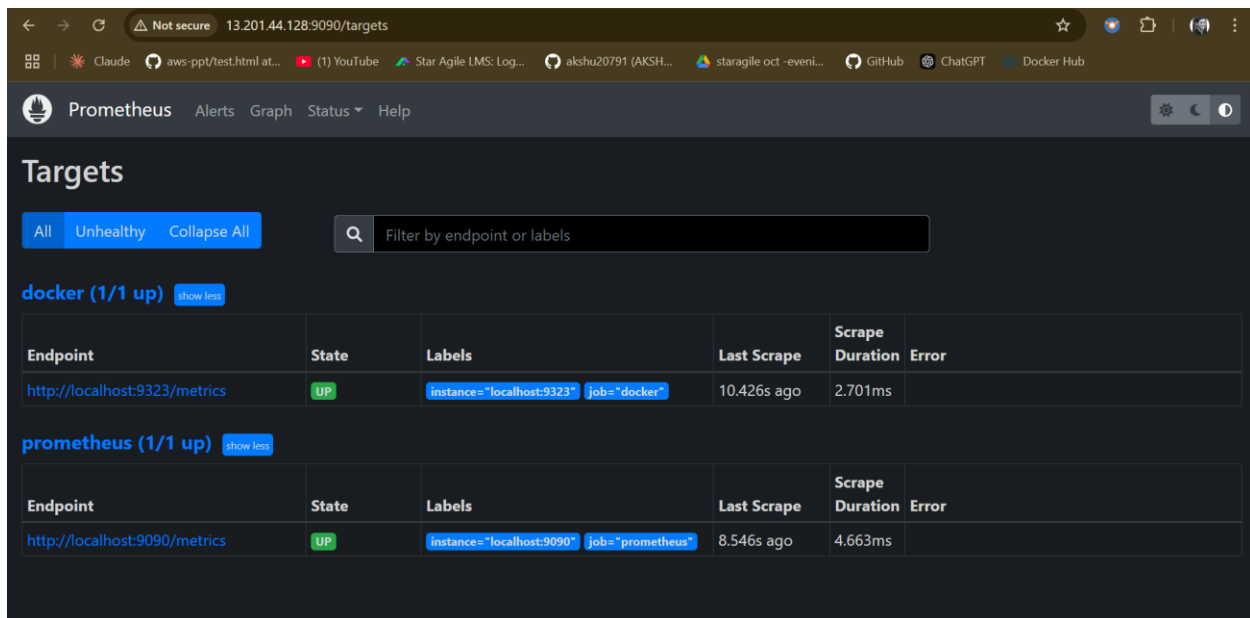
Step-15 Access Grafana through public ip and login as admin



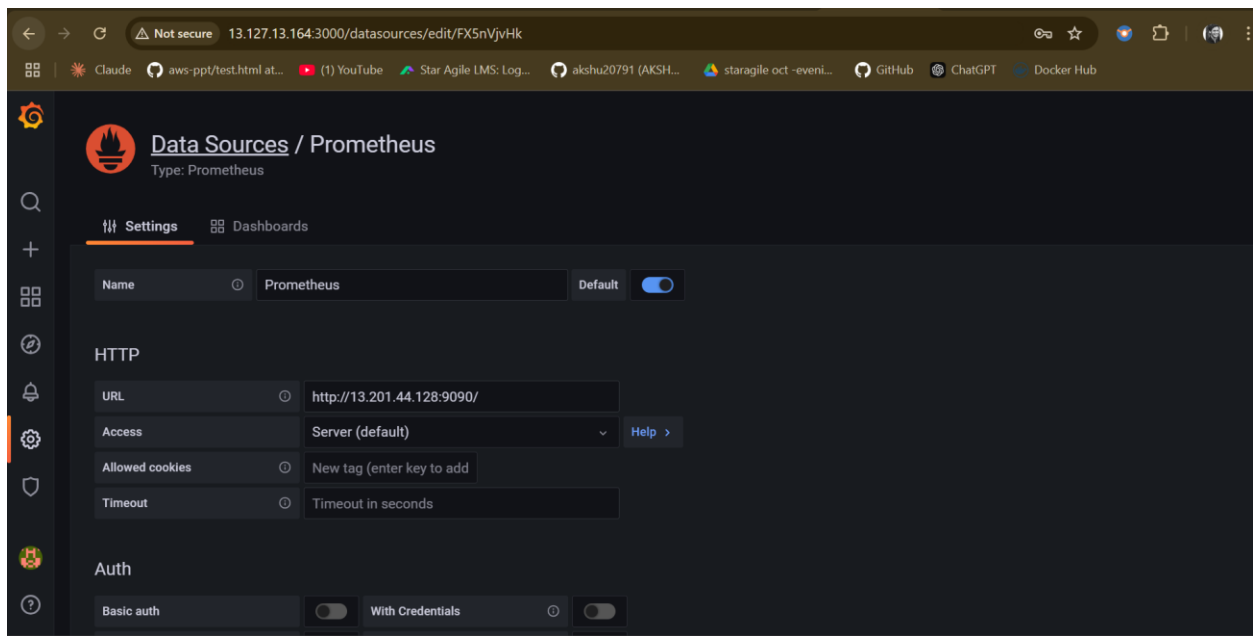
Step-16 Install Prometheus in master server



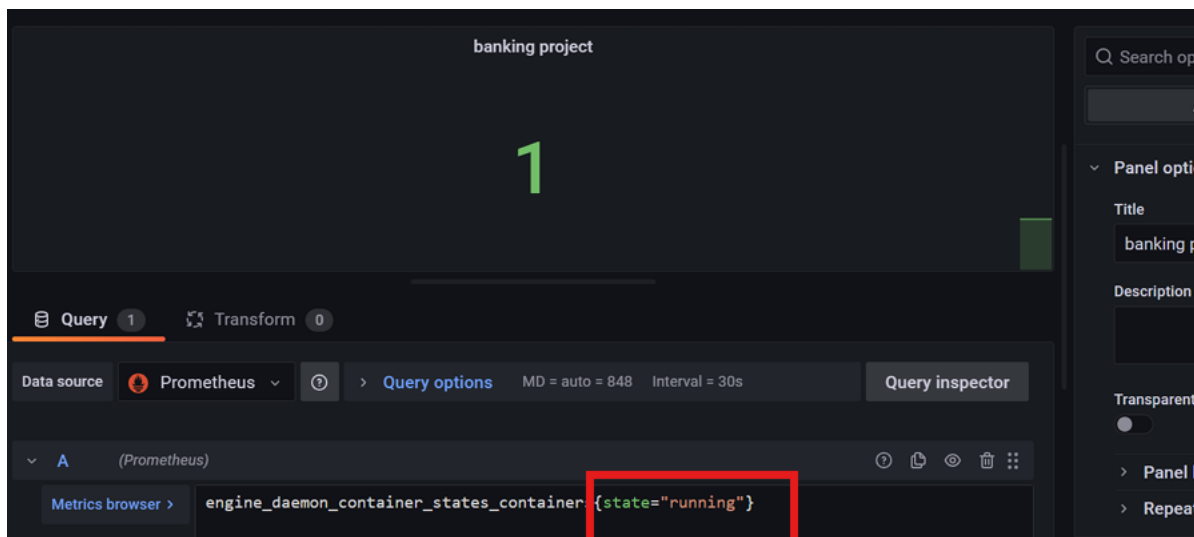
Step-17 See the target in Prometheus

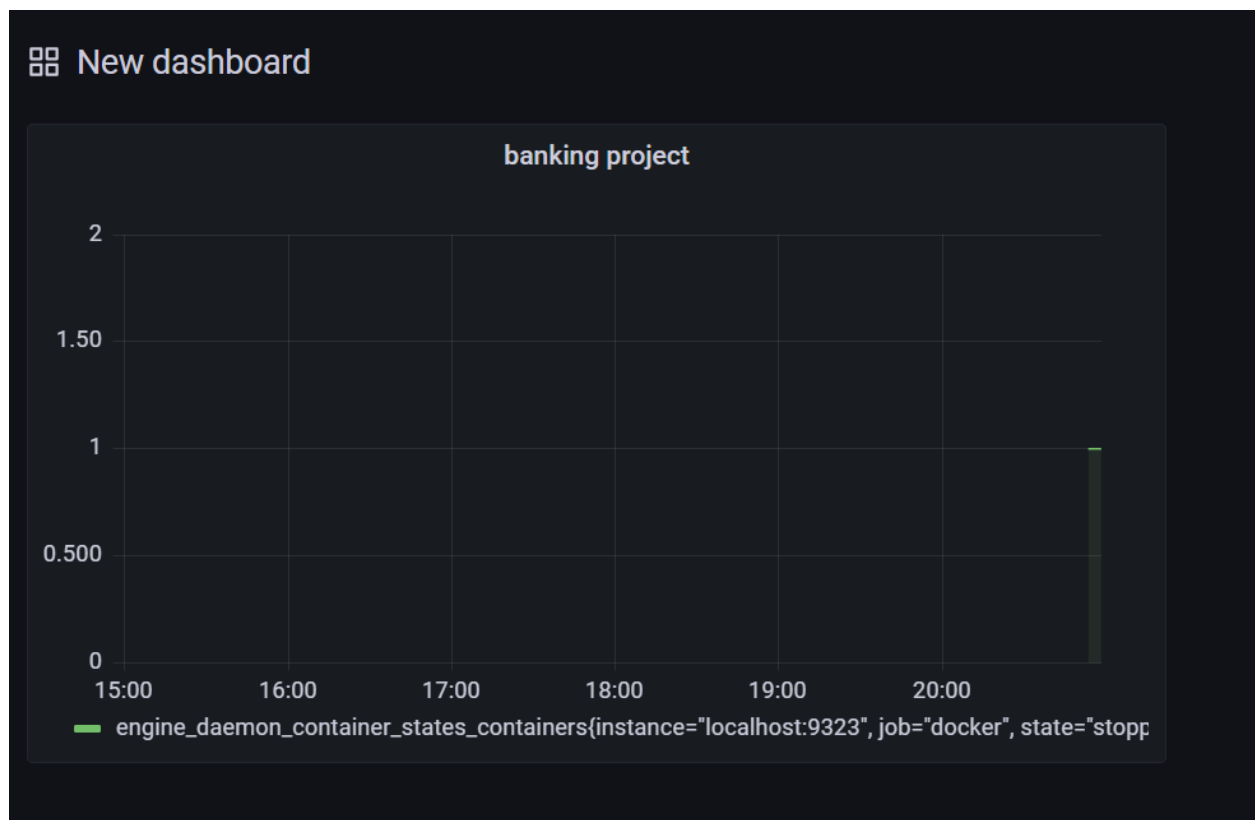
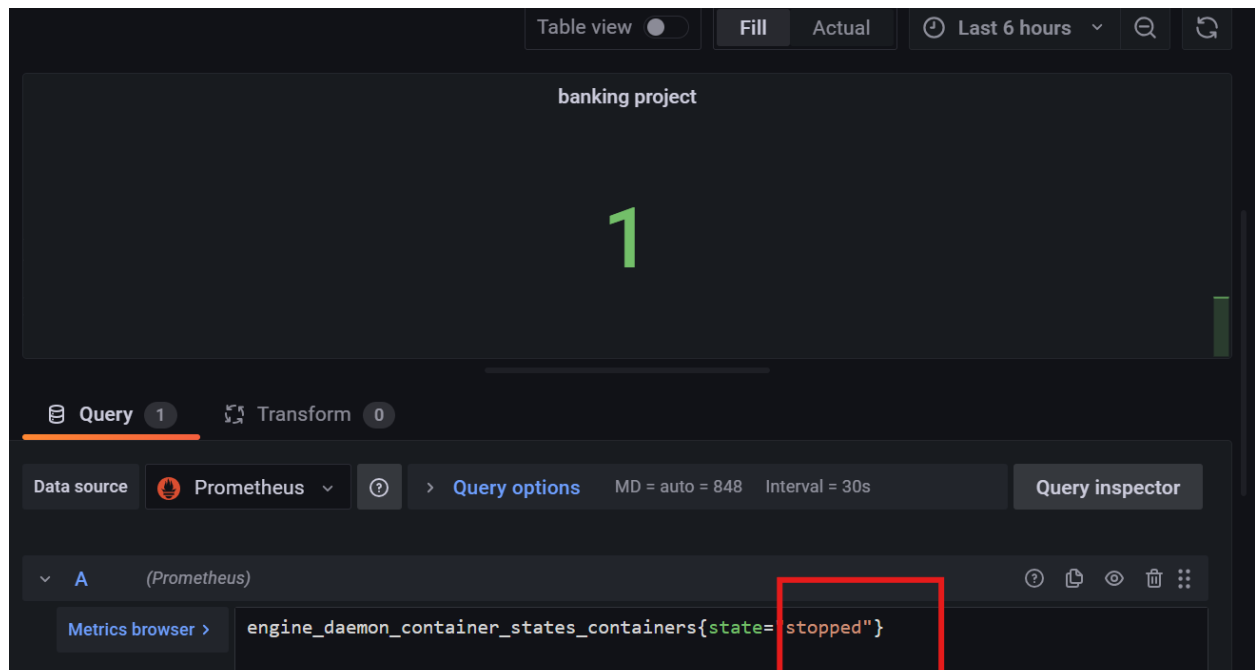


Step-18 Paste the Prometheus ip in Grafana



Step-19 Go to the dashboard see the running container status





Finance Me Banking and Finance Domain project done